

Institute Management System

Introduction

Institute Management System is a platform designed to manage students, teachers, courses, attendance, exams, fees, and notices. The system is centralized with the **Admin** having full control over all functionalities. **Students** and **Teachers** have more restricted access, primarily only limited to viewing data and updating their own profiles. The system incorporates role-based access control and session management.

Scope of Project

The project scope includes the development of a web-based Institute Management System with dedicated modules for Admin, Student, and Teacher roles. The system will handle all core functionalities outlined in the requirements, including user management, course and timetable management, attendance tracking, fee and exam result management, and a notice board. The focus is on a structured, secure, and user-friendly interface.

Software Requirements Specification (SRS)

1) User Roles And Authentication

- **User Roles:** The system defines three main user roles: Admin, Teacher, and Student.
- **Admin Accounts:** There is one pre-created Admin account hardcoded into the database.
- **Teacher And Student Accounts:** Both teachers and students have their own unique login credentials.
- **Student Registration:** Student registration is only allowed if the student is already enrolled in the institute, and this is verified using their email and phone number.
- **Authentication And Session Management:** The system uses role-based access control. It also includes a login/logout feature with session management, and sessions will expire after a user logs out.

Admin Use Case Diagram:

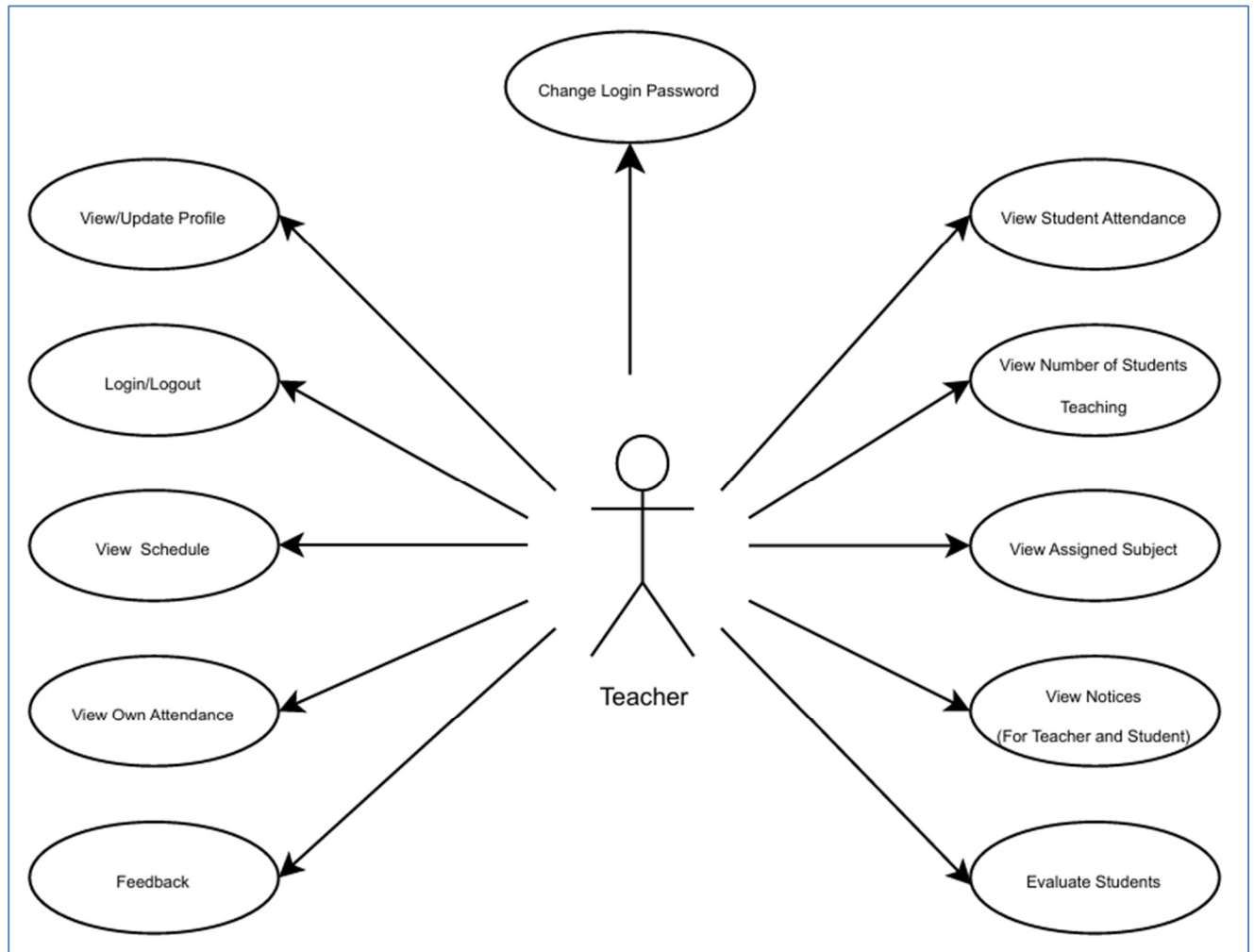


2) Admin Functionalities

The Admin has a dashboard providing an overview of the system, including the number of students, teachers, and courses. Key functionalities for the Admin include:

- **Manage Students:** The Admin can view, add, and update student information. Students can be marked as inactive instead of being permanently deleted. The Admin can also view a student's profile, including attendance, fees, and marks.
- **Manage Teachers:** The Admin can view, add, and update teacher details. Teachers can also be marked as inactive.
- **Manage Courses & Subjects:** This involves creating, updating, and viewing courses. Admins can assign teachers and students to specific courses and subjects. Courses can be marked as inactive rather than being deleted.
- **Manage Attendance:** The Admin can add and edit attendance records for both students and teachers. They can view total working days, total present days, and total absent days, and can filter the results by course or student name.
- **Fee Management:** The Admin can create fees per course and update the payment status for individual students (Paid/Unpaid/Pending). They can view and search student fee records and send fee reminders (an optional feature).
- **Exam and Result Management:** The Admin can upload exam schedules, which include subject, course, date, and total marks. They can also upload student exam results (marks) per subject and send result notifications to parents (optional).
- **Timetable Management:** Admins can create, view, update, and delete class schedules by uploading a PDF file.
- **Notice Board:** The Admin can create and view announcements that are visible to both teachers and students after they log in. These announcements can be for students, teachers, or both.

Teacher Use Case Diagram:

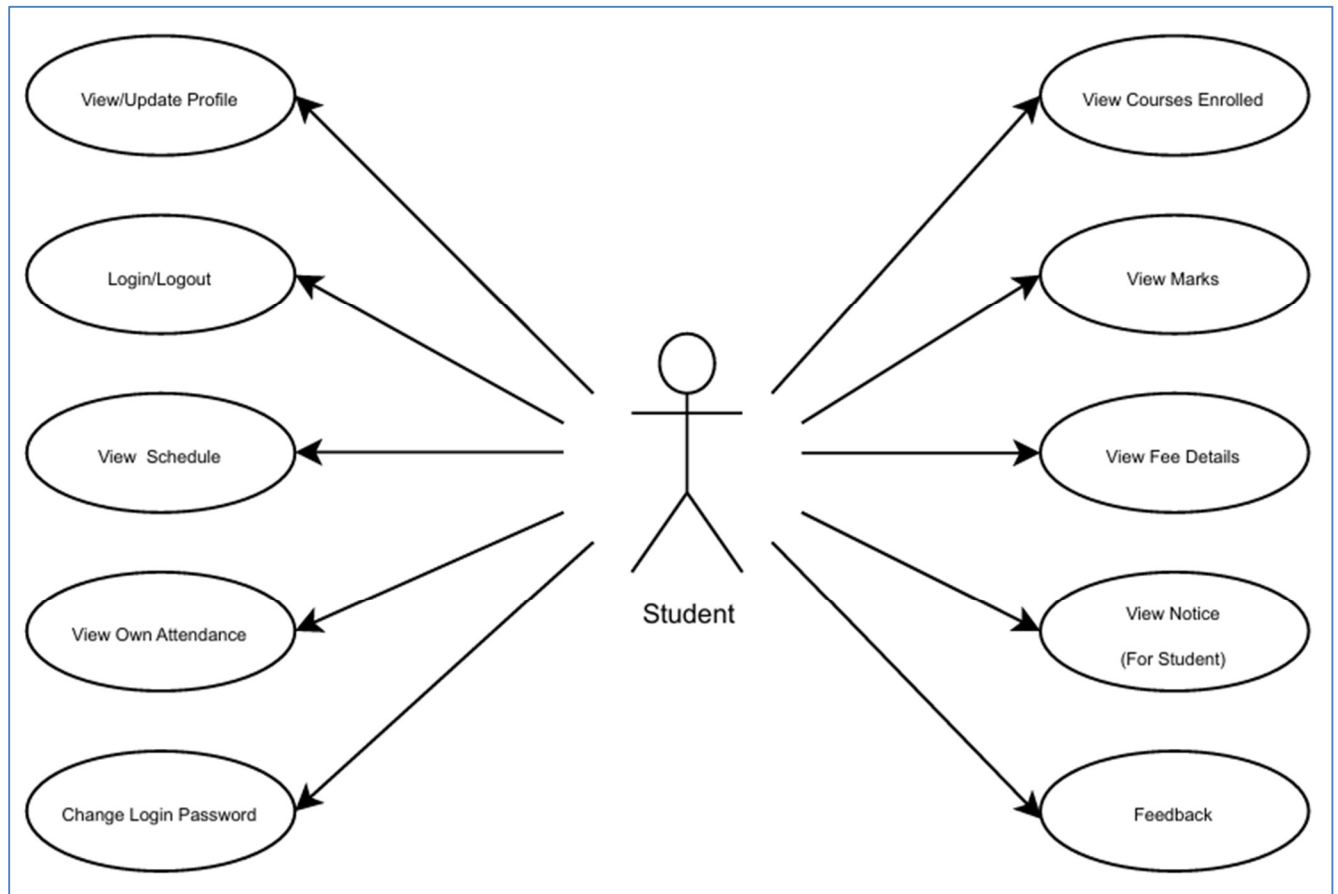


3) Teacher Functionalities

A teacher's dashboard displays relevant information such as notices from the Admin, the number of students they teach, and the subjects they are assigned.

- **Dashboard G Profile:** Teachers can view their profile and announcements. They can also view the total number of students they teach and the subjects they teach.
- **Personal Information:** Teachers can view their personal timetable and attendance.
- **Student Management:** Teachers can view student attendance records for the classes they are assigned.
- **Profile Management:** Teachers can update their own profile.

Student Use Case Diagram:



4) Student Functionalities

The student's dashboard provides access to their personal information and academic data.

- **Dashboard G Profile:** Students can view their profile, which includes their photo, name, roll number, course, and batch. They can also view their timetable, monthly attendance summary, upcoming exams, past results, and fee payment status.
- **Profile Management:** Students are able to update their own profile information, such as their parent's contact details, address, and password.
- **Communication:** Students can view important announcements posted on the notice board. Additionally, they can raise complaints or queries directly through the system to communicate issues or concerns to the administration.

5) Other Requirements

➤ Password Encryption

- User passwords for Admins, Teachers, and Students are securely hashed using BCrypt before storing in the logins table.
- BCrypt automatically adds a salt and provides a configurable work factor, ensuring strong protection against brute-force and rainbow table attacks.
- Plain-text passwords are never stored in the database.

➤ Security:

- Authentication and Role Management
- Users log in using their username and password.
- The role field (Admin, Teacher, Student) determines access to specific functionalities:
- Admin: Can manage users, courses, attendance, fees, etc.
- Teacher: Can manage student attendance, view student performance.
- Student: Can view their own attendance, marks, and fees.

➤ Session Management

- After successful login, a session is created for the user.
- Sessions expire automatically on logout to prevent unauthorized access.
- Session timeouts can be configured to expire after a period of inactivity, enhancing security.

➤ Database-Level Security

- All sensitive relationships (like attendance, marks, fees) use foreign keys for integrity.
- is_inactive and deleted_at fields ensure soft deletion without exposing historical sensitive data.

➤ Additional Measures

- Input validation is applied on all forms to prevent SQL injection.
- Password reset functionality can be implemented using secure token-based links.

➤ Error Handling

- The system should provide friendly error messages, such as "Invalid login" or "Student not found".